

Lesson 5-9: Area of Regular Polygons

Grade 6 Mathematics · Standard 6.GR.1

Name: _____ Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINT

A regular hexagon is divided into 6 equal triangles from the center. Each triangle has a base of 6 ft and a height of 5.2 ft. What is the area of one triangle?

- ☐ A 11.2 sq ft
- ☐ B 15.6 sq ft
- ☐ C 15.6 ft
- ☐ D 31.2 sq ft

Question 2 SELECTED RESPONSE · 1 POINT

Using the triangle from the previous question, what is the total area of the hexagonal skylight?

- ☐ A 15.6 sq ft
- ☐ B 62.4 sq ft
- ☐ C 93.6 sq ft
- ☐ D 187.2 sq ft

Question 3 SELECTED RESPONSE · 1 POINT

A regular hexagon is split into 6 triangles, each with base 4 cm and height 3.5 cm. What is the total area?

- ☐ A 14 sq cm
- ☐ B 21 sq cm
- ☐ C 42 sq cm
- ☐ D 84 sq cm

Question 4 SELECTED RESPONSE · 1 POINT

How many triangles can you make from the center of a regular pentagon?

- ☐ A 3
- ☐ B 5
- ☐ C 6
- ☐ D 10

Question 5 SELECTED RESPONSE · 1 POINT

A regular hexagonal floor tile splits from its center into 6 congruent triangles, each with an area of 12 cm^2 . What is the total area of the tile?

- ☐ A 12 cm^2
- ☐ B 18 cm^2
- ☐ C 36 cm^2
- ☐ D 72 cm^2

Question 6 SELECTED RESPONSE · 1 POINT

A regular pentagonal tile splits into 5 congruent triangles. Each triangle has a base of 4 in and a height of 6 in. What is the total area of the tile?

- ☐ A 12 in^2
- ☐ B 24 in^2
- ☐ C 60 in^2
- ☐ D 120 in^2

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

A regular hexagon is decomposed from its center into 6 identical triangles. Each triangle has a base of 8 cm and a height of 7 cm. What is the area of the hexagon?

Fill in the bubble next to the one best answer.

- ☐ (A) 28 square centimeters
- ☐ (B) 56 square centimeters
- ☐ (C) 168 square centimeters
- ☐ (D) 336 square centimeters

Part B

Answer Part A before Part B — Part B asks about your reasoning.

Why can a regular polygon be decomposed into identical triangles from its center?

- ☐ (A) Every polygon can be split into identical triangles.
- ☐ (B) Triangles are the only shape with a known area formula.
- ☐ (C) All its sides are equal and all its angles are equal, so each center-to-side triangle is congruent.
- ☐ (D) The center of a polygon is always a vertex.

Question 8**SELECT ALL THAT APPLY**

Select ALL statements that are true about the area of a regular polygon:

Mark the box next to EVERY answer that is true. More than one is correct.

- ☐ (A) It equals the number of sides times the area of one center triangle.
- ☐ (B) Each center triangle has the same height for a regular polygon.
- ☐ (C) Only regular polygons with an even number of sides can be decomposed this way.
- ☐ (D) A regular polygon has all sides equal and all angles equal.
- ☐ (E) The height of each center triangle is measured perpendicular to a side.

Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

Question 9

WRITTEN RESPONSE · 2 POINTS

A regular hexagon and a regular pentagon both have triangles with the same base (6 ft) and the same height (5 ft). Which polygon has the greater total area? Explain why the number of sides matters.

Write your answer in complete sentences. Show or explain your mathematical thinking.

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.